NARENDRA SINGH YADAV

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PROFESSIONAL SUMMARY

B.Tech Electrical & Electronics Engineering student at IPS Academy, Indore (1st Rank in department, 2025-26) with strong skills in software development, AI and problem solving. Built full-stack and AI systems including a real-time voice AI assistant, an offline digital-forensics workstation and a rule-based hospital bill auditing platform using Python, FastAPI, React and TypeScript. Participated in national and international hackathons focused on software, AI and data-driven solutions.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL
Frontend: React 19, TanStack Start, TanStack Router, Tailwind CSS, Recharts, Cytoscape.js, Web Audio API
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, Pydantic, SQLAlchemy, Alembic
AI & Data: LLM Integration (Gemini, Groq), RAG, Tool Calling, OCR, PyMuPDF, Pandas, NetworkX, Speech (STT, TTS, VAD)
Databases: PostgreSQL, Supabase, SQLite
Tools: Git, GitHub, Linux, Docker, AWS (learning), Pytest, Vitest, Vite, Postman, VS Code
Concepts: OOP, Data Structures & Algorithms, Rule Engines, REST Architecture, Row Level Security, SHA-256 Hashing

EDUCATION

Bachelor of Technology, Electrical & Electronics Engineering *Jul 2025 - Jul 2029*
IPS Academy, Institute of Engineering & Science, Indore (Affiliated to RGPV)

- 1st Rank in EEE Department, 1st Year (Session 2025-26); received Academic Award Certificate and Trophy

Class XII (MP Board), Govt. H.S. Excellence School No. 1, Tikamgarh, M.P. | 93.6% *2024*

Class X (MP Board), Late Pt. Shri G.L. Vyas Creative Convent H.S. School, Tikamgarh, M.P. | 94.8% *2022*

EXPERIENCE

AWS with DevOps Intern, EWB Edutech Pvt. Ltd. (Virtual) *Oct 2026 - Nov 2026*

- Selected via the AICTE National Internship Portal; attending mentor-led sessions on AWS and DevOps, completing weekly assessments, assigned tasks and a final mini project.

PROJECTS

BillShielder: AI-Powered Hospital Bill Auditor *FastAPI, React 19, Python*

- Built a full-stack platform that audits Indian hospital bills against 37 codified rules drawn from IRDAI, NPPA/DPCO, GST Council and CGHS/PM-JAY regulations.
- Developed a deterministic rule engine detecting arithmetic errors, duplicate charges, ICU unbundling and non-payable items, with statutory citations instead of a black-box score.
- Implemented PyMuPDF-based PDF/image ingestion with Indian currency parsing; designed pluggable OCR (Tesseract, Google Document AI), LLM (Gemini, OpenAI) and storage (local, S3) providers.
- Added Gemini explainable chat with offline fallback, dispute-letter generation, a React dashboard and 30+ automated tests.

VoiceForge: Real-Time Voice AI System *Python, Flask, Groq, Rime AI, Web Audio API*

- Built an interruption-resilient real-time voice assistant with generation control to prevent response races.
- Implemented request cancellation, audio-buffer teardown, tool calling and modular RAG; 40/40 test cases passing.
- Developed an observability dashboard for mic, VAD, STT, LLM, TTS and playback stages. DataForge 2026 entry.

KAVACH: Offline Cyber Fraud Triage & Forensics System *Python, FastAPI, React, TypeScript, SQLite, NetworkX*

- Built an offline forensic workstation correlating CDR, IPDR, Bank/UPI, email and APK evidence for cyber-fraud investigation.
- Implemented a SHA-256 hash-chained custody log for tamper-evident evidence and a mule-account fraud graph.
- Added explainable risk scoring, entity resolution and CGNAT-aware correlation safeguards.

Emberhold: Full-Stack Life RPG Platform *React, TypeScript, TanStack Start, Supabase, PostgreSQL*

- Developed a productivity platform converting real-world tasks into quests, with server-side progression and streak multipliers.
- Used Supabase Row Level Security for per-user data isolation; built an in-browser audio system with Web Audio API.

HACKATHONS, EVENTS & ACHIEVEMENTS

- DataForge 2026, KDAG, IIT Kharagpur: built VoiceForge, a real-time voice AI system
- Hack Devengers 2.0: Certificate of Appreciation for KAVACH
- INNOVIK 6.0, International Hackathon 2026: participated in the 24-hour Agentic AI challenge
- Eureka! 2026, E-Cell, IIT Bombay: participated in the national business model competition
- MP Semiconductor Summit 2026, Govt. of Madhya Pradesh (Indore, 23-24 Oct 2026): attended on invitation

CERTIFICATIONS & RESPONSIBILITIES

- Programming in Modern C++: 12-week NPTEL / SWAYAM course; successfully completed and passed the certification exam
- ISRO IIRS Training: UAV Sensors Applications in Forestry and Ecology (22 Oct 2026); Overview of Geocomputation and Geo-web Services (9-13 Nov 2026, ongoing)
- Campus Ambassador, Persevex (Jul 2026): campus outreach, promoting programs and driving student registrations